

Dataset	Model	Uncalibrated	Hist. Binning	Isotonic	BBQ	Temp. Scaling	Vector Scaling	Matrix Scaling
Birds	ResNet 50	9.19%	4.34%	5.22%	4.12%	1.85%	3.0%	21.13%
Cars	ResNet 50	4.3%	1.74%	4.29%	1.84%	2.35%	2.37%	10.5%
CIFAR-10	ResNet 110	4.6%	0.58%	0.81%	0.54%	0.83%	0.88%	1.0%
CIFAR-10	ResNet 110 (SD)	4.12%	0.67%	1.11%	0.9%	0.6%	0.64%	0.72%
CIFAR-10	Wide ResNet 32	4.52%	0.72%	1.08%	0.74%	0.54%	0.6%	0.72%
CIFAR-10	DenseNet 40	3.28%	0.44%	0.61%	0.81%	0.33%	0.41%	0.41%
CIFAR-10	LeNet 5	3.02%	1.56%	1.85%	1.59%	0.93%	1.15%	1.16%
CIFAR-100	ResNet 110	16.53%	2.66%	4.99%	5.46%	1.26%	1.32%	25.49%
CIFAR-100	ResNet 110 (SD)	12.67%	2.46%	4.16%	3.58%	0.96%	0.9%	20.09%
CIFAR-100	Wide ResNet 32	15.0%	3.01%	5.85%	5.77%	2.32%	2.57%	24.44%
CIFAR-100	DenseNet 40	10.37%	2.68%	4.51%	3.59%	1.18%	1.09%	21.87%
CIFAR-100	LeNet 5	4.85%	6.48%	2.35%	3.77%	2.02%	2.09%	13.24%
ImageNet	DenseNet 161	6.28%	4.52%	5.18%	3.51%	1.99%	2.24%	-
ImageNet	ResNet 152	5.48%	4.36%	4.77%	3.56%	1.86%	2.23%	-
SVHN	ResNet 152 (SD)	0.44%	0.14%	0.28%	0.22%	0.17%	0.27%	0.17%
20 News	DAN 3	8.02%	3.6%	5.52%	4.98%	4.11%	4.61%	9.1%
Reuters	DAN 3	0.85%	1.75%	1.15%	0.97%	0.91%	0.66%	1.58%
SST Binary	TreeLSTM	6.63%	1.93%	1.65%	2.27%	1.84%	1.84%	1.84%
SST Fine Grained	TreeLSTM	6.71%	2.09%	1.65%	2.61%	2.56%	2.98%	2.39%

Table 1. ECE (%) (with $M = 15$ bins) on standard vision and NLP datasets before calibration and with various calibration methods. The number following a model’s name denotes the network depth.

Extension of binning methods. One common way of extending binary calibration methods to the multiclass setting is by treating the problem as K one-versus-all problems (Zadrozny & Elkan, 2002). For $k = 1, \dots, K$, we form a binary calibration problem where the label is $\mathbf{1}(y_i = k)$ and the predicted probability is $\sigma_{\text{SM}}(\mathbf{z}_i)^{(k)}$. This gives us K calibration models, each for a particular class. At test time, we obtain an unnormalized probability vector $[\hat{q}_i^{(1)}, \dots, \hat{q}_i^{(K)}]$, where $\hat{q}_i^{(k)}$ is the calibrated probability for class k . The new class prediction $\hat{y}'_i$ is the argmax of the vector, and the new confidence $\hat{q}'_i$ is the max of the vector normalized by $\sum_{k=1}^K \hat{q}_i^{(k)}$. This extension can be applied to histogram binning, isotonic regression, and BBQ.

Matrix and vector scaling are two multi-class extensions of Platt scaling. Let $\mathbf{z}_i$ be the *logits vector* produced before the softmax layer for input $\mathbf{x}_i$. *Matrix scaling applies* a linear transformation $\mathbf{W}\mathbf{z}_i + \mathbf{b}$ to the logits:

$$\begin{aligned}\hat{q}_i &= \max_k \sigma_{\text{SM}}(\mathbf{W}\mathbf{z}_i + \mathbf{b})^{(k)}, \\ \hat{y}'_i &= \operatorname{argmax}_k (\mathbf{W}\mathbf{z}_i + \mathbf{b})^{(k)}.\end{aligned}\tag{8}$$

The parameters $\mathbf{W}$ and $\mathbf{b}$ are optimized with respect to NLL on the validation set. As the number of parameters for matrix scaling grows quadratically with the number of classes K , we define *vector scaling* as a variant where $\mathbf{W}$ is restricted to be a diagonal matrix.

Temperature scaling, the simplest extension of Platt scaling, uses a single scalar parameter $T > 0$ for all classes. Given the logit vector $\mathbf{z}_i$, the new confidence prediction is

$$\hat{q}_i = \max_k \sigma_{\text{SM}}(\mathbf{z}_i/T)^{(k)}.\tag{9}$$

T is called the temperature, and it “softens” the softmax (i.e. raises the output entropy) with $T > 1$. As $T \rightarrow \infty$, the probability $\hat{q}_i$ approaches $1/K$, which represents maximum uncertainty. With $T = 1$, we recover the original probability $\hat{p}_i$. As $T \rightarrow 0$, the probability collapses to a point mass (i.e. $\hat{q}_i = 1$). T is optimized with respect to NLL on the validation set. Because the parameter T does not change the maximum of the softmax function, the class prediction $\hat{y}'_i$ remains unchanged. In other words, *temperature scaling does not affect the model’s accuracy*.

Temperature scaling is commonly used in settings such as knowledge distillation (Hinton et al., 2015) and statistical mechanics (Jaynes, 1957). To the best of our knowledge, we are not aware of any prior use in the context of calibrating probabilistic models.³ The model is equivalent to maximizing the entropy of the output probability distribution subject to certain constraints on the logits (see Section S2).

4.3. Other Related Works

Calibration and confidence scores have been studied in various contexts in recent years. Kuleshov & Ermon (2016) study the problem of calibration in the online setting, where the inputs can come from a potentially adversarial source. Kuleshov & Liang (2015) investigate how to produce calibrated probabilities when the output space is a structured object. Lakshminarayanan et al. (2016) use ensembles of networks to obtain uncertainty estimates. Pereyra et al. (2017) penalize overconfident predictions as a form of regularization. Hendrycks & Gimpel (2017) use confidence

³To highlight the connection with prior works we define temperature scaling in terms of $\frac{1}{T}$ instead of a multiplicative scalar.